Section A: Pure Mathematics [35 marks]

1 Show that the line $x+y=q$ will intersect the curve $x^2-2x+2y^2=3$ at two distinct points if $q^2<2q+5$. [4]

2 A bowl is filled with water and when the depth of the water is h cm, the volume of water is $2\pi h^2\ \text{cm}^3$. Initially the bowl is empty and it is then filled with water at a steady rate of $3\ \text{cm}^3\text{s}^{-1}$. Determine the rate at which h is increasing at the instant when the time is 6 s. [4]

3 **(i)** Find the value of a such that $\int_a^1 e^{1-x}\ \mathrm{d}x = e^2-1$. [3]

(ii) Express $\dfrac{x}{2-x}$ in the form $a+\dfrac{b}{2-x}$ where a and b are integers. [1]

(iii) Find $\displaystyle\int\left(\frac{x}{2-x}\right)^2 \mathrm{d}x$. [3]

4 **(i)** On a single diagram, sketch the graphs of $y=\ln(x-1)$ and $y=\dfrac{1}{x-1}$ for $x>1$, stating clearly the coordinates of any points of intersection with the axes and equation of any asymptotes. [3]

(ii) Find the x-coordinate of the point of intersection of $y=\ln(x-1)$ and $y=\dfrac{1}{x-1}$, giving your answer correct to 4 decimal places. [1]

(iii) Write down as an integral, an expression for the area of the region bounded by the curves $y=\ln(x-1)$, $y=\dfrac{1}{x-1}$ and the lines $x=2$ and $x=4$.

Evaluate this integral, giving your answer correct to 2 decimal places. [2]

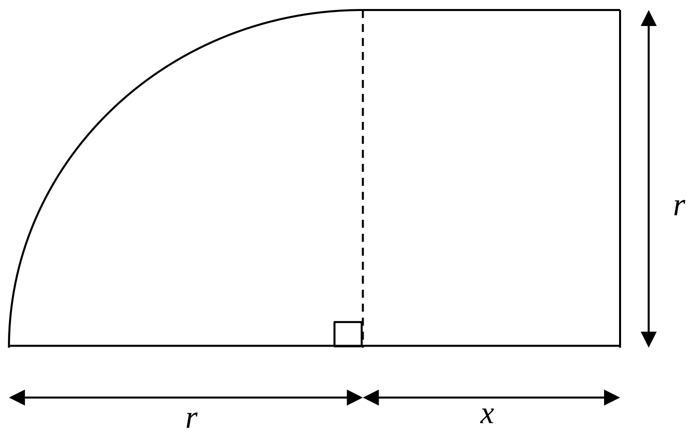
5 The equation of a curve C is $\frac{1}{x}+\frac{1}{y}=\frac{1}{k}$, where k is non-zero.

(i) Express y in terms of x and k. [1]

(ii) Find the equation of the normal to the curve C at $(2k, 2k)$. [4]

(iii) Show that the normal found in **(ii)** will not cut the curve C again for all real values of k. [3]

6 A field is made up of a quadrant of radius *r* cm and a rectangle of width *x* cm and height *r* cm, as shown in the diagram.



(i) The perimeter and area of the field are *P* and *A* respectively. Given that *P* is fixed at 200 cm, show that $A=100r-r^2$. [2]

(ii) Without the use of graphing calculator, find the value of *x* when *A* is a maximum and find the maximum value of *A*. [4]

Section B: Statistics [60 marks]

7 A mobile phone manufacturer tests the quality of its products by taking a sample of the mobile phones.

(i) Why does the mobile phone manufacturer take a sample? [1]

(ii) If 3 different phone models are produced in the ratio 3:2:1, describe how the manufacturer can select a stratified sample if it chose to test 120 mobile phones. [2]

8 During an epidemic of a certain disease, a doctor is consulted by 120 patients who display symptoms of the disease. Of the 120 patients, 75 are male of whom 15 have actually contracted the disease. Twenty female patients contracted the disease.

(i) A patient is selected at random. The events *A* and *B* are defined as follows:

A = the person is female

B = the person has contracted the disease.

Evaluate

(a) $\mathrm{P}(A)$ [1]

(b) $\mathrm{P}(A \cup B)$ [1]

(c) $\mathrm{P}(A \cap B)$ [1]

(d) $\mathrm{P}(A|B)$ [2]

(ii) If two different patients are selected at random without replacement, find the probability that

(a) both have the disease, [1]

(b) exactly one of them have the disease, [2]

(c) one is a female with the disease and the other is a male without the disease. [2]

(iii) Of the patients with the disease, 96% react positively to a test for diagnosing the disease. 8% of the patients without the disease also tested positive for the test.

Find the probability of a person selected at random who

(a) tests positive for the test, [2]

(b) has the disease, given that he or she has tested positive for the test. [2]

9 A precision machine is set to produce metal rods with lengths which follow a normal distribution. The mean and standard deviation of the length of the rods is 2 cm and 0.28 cm respectively. A sample of 150 of these rods is randomly selected from the production of this machine and found to have a mean of 1.97 cm.

(i) Test at the 5% level of significance whether the mean length of rods produced by the machine differs from 2 cm. [4]

(ii) Explain how your conclusion in (i) would be affected if you subsequently discovered that

(a) the sample was not random, [1]

(b) the distribution of the length of rods was not normal. [2]

(iii) A test on another random sample of 150 rods give a mean length of L cm. Find the set of values of L if a test at the 5% level of significance indicates that the mean length of this batch differs from 2 cm. [2]

10 Pens manufactured by Sailor Corporation are packed in boxes of 250. On average, 3% of pens manufactured are defective. A box which contains 10 or more defective pens is rejected.

A box of pens is selected at random.

(i) Find the probability that the box contain at least 5 defective pen. [2]

(ii) Using a suitable approximation, find the probability that the box will be rejected, leaving your answer to 5 significant figures. [3]

A delivery van carries 100 such boxes in a trip.

(iii) Find the expected number of rejected boxes per delivery, leaving your answer to the nearest integer. [2]

Over a one-year period, the van will make 300 deliveries, each trip carrying 100 such boxes.

(iv) Using a suitable approximation, find the probability that the mean number of rejected boxes per trip for the year is less than 23. [3]

11 The following table shows the values of 4 variables W , X , Y and Z in 8 different areas of a country, where

W is the percentage of household living in private properties.

X is the number of residents (in thousands) with tertiary educational qualifications.

Y is the percentage of the population with tertiary educational qualifications.

Z is the population (in thousands).

Area	1	2	3	4	5	6	7	8
w	51	32	49	33	73	57	52	77
x	11.9	37.0	43.7	10.6	36.4	8.2	15.3	28.6
y	8	y_2	y_3	y_4	y_5	y_6	y_7	y_8
z	150	746	662	345	239	108	210	223

(i) Calculate the values of y_2 to y_8, giving your answers to the nearest integer. [3]

(ii) Estimate the value of w when $x = 18$, leaving your answer to the nearest integer. State the equation of the least squares regression line used. [2]

(iii) Calculate the product moment correlation coefficient between w and x. Comment on the reliability of the estimate obtained in **(ii)**. [2]

(iv) A research assistant claimed that the product moment correlation coefficient between w and x must be the same as the product moment correlation coefficient between w and y as the product moment correlation coefficient is unaffected by scaling. Comment on the validity of this statement. [1]

(v) Using the answers in **(i)**, calculate the product moment correlation coefficient between w and y, and explain whether your answer suggests that a linear model is appropriate. [2]

(vi) Discuss why the use of data from related variables (X and Y) can lead to different product moment correlation coefficient as obtained in **(iii)** and **(v)**. [1]

(vii) Can we conclude that a person living in a private property is a result of this person's possession of a tertiary educational qualification? Justify your answer. [1]

12 The diameters of footballs manufactured by a large factory are normally distributed with mean 22 cm and standard deviation of 0.3 cm.

(i) Three footballs were selected at random. What is the probability that one of them has a diameter between 21.7cm and 22.3cm and the other two footballs have diameter less than the mean. [3]

(ii) Find the value of *a* such that 75% of the footballs have diameters between $(22-a)$ cm and $(22+a)$ cm. [2]

The diameters of basketballs manufactured by the same factory are also normally distributed with mean 24 cm and standard deviation of 0.4 cm.

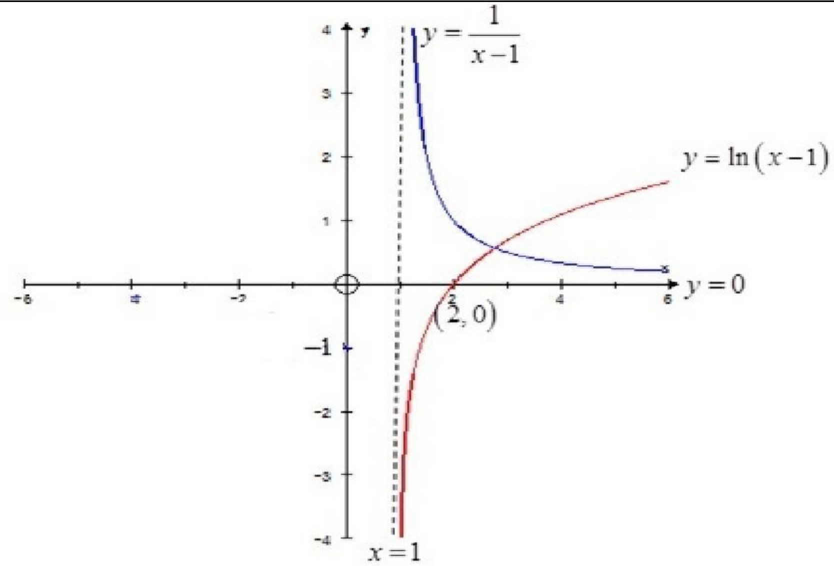
(iii) Find the probability that the difference between the average diameter of 10 footballs and the average diameter of 5 basketballs is less than 1.5 cm. [3]

(iv) State an assumption used in **(iii)**. [1]

The basketballs manufactured were then tested on a circular basketball hoop with a circumference of 78 cm.

(v) Find the probability that a randomly chosen basketball will be able to pass through the basketball hoop. [3]

Qn	Solution	Marks
1	Subst. $y=q-x$ into $x^2-2x+2y^2=3$ and get	
	$x^2-2x+2(q-x)^2=3 \Rightarrow x^2-2x+2\left(q^2-2qx+x^2\right)=3$	[M1]
	$\Rightarrow 3x^2-(2+4q)x+2q^2-3=0$	[B1]
	Discriminant	
	$=(2+4q)^2-4(3)\left(2q^2-3\right)$	[M1]
	$=\left(4+16q+16q^2\right)-\left(24q^2-36\right)$	
	$=40+16q-8q^2$	
	$=8(5+2q-q^2)$	
	If , $q^2<2q+5$ then the discrimant > 0 and the two graphs intersect in two distinct points.	[B1]
2	$V=2\pi h^2 \Rightarrow \dfrac{\mathrm{d}V}{\mathrm{d}h}=4\pi h$	
	Given $\dfrac{\mathrm{d}V}{\mathrm{d}t}=3$ (constant) , after 6 s, $V=3\times 6=18$	[B1]
	$\therefore 2\pi h^2=18 \Rightarrow h=\sqrt{\dfrac{18}{2\pi}}=1.692568751$	[B1]
	$\therefore \dfrac{\mathrm{d}V}{\mathrm{d}t}=\dfrac{\mathrm{d}V}{\mathrm{d}h}\times\dfrac{\mathrm{d}h}{\mathrm{d}t}$	
	$3=4\pi h\times\dfrac{\mathrm{d}h}{\mathrm{d}t}$	[B1]
	$\dfrac{\mathrm{d}h}{\mathrm{d}t}=\dfrac{3}{4\pi h}=\dfrac{3}{4\pi(1.692568751)}=0.141$	[A1]
	$\therefore$ the rate at which h is increasing after 6 s is 0.141 cms^{-1} .	
3(i)	$\int_a^1 e^{1-x}\,\mathrm{d}x=\left[-e^{1-x}\right]_a^1=e^{1-a}-1$	[B1,B1]
	$e^{1-a}=e^2$	
	$a=-1$	[A1]

(ii)	$\frac{x}{2-x}=\frac{-(2-x)+2}{2-x}=-1+\frac{2}{2-x}$	[B1]
(iii)	$\int\left(\frac{x}{2-x}\right)^2\,\mathrm{d}x=\int\left(-1+\frac{2}{2-x}\right)^2\mathrm{d}x$ $=\int 1-\frac{4}{2-x}+\frac{4}{(2-x)^2}\,\mathrm{d}x$ $=x+4\ln(2-x)+\frac{4}{2-x}+c$	[B1] [B1 & B1 for $4\ln(2-x)$, $\frac{4}{2-x}$ respectively]
4(i)		[1-intercepts, 1- asy, 1- curve]
(ii)	$x=2.7632$	[A1]
(iii)	$\int_2^{2.7632}\frac{1}{x-1}-\ln(x-1)\,\mathrm{d}x\;+\int_{2.7632}^4\ln(x-1)-\frac{1}{x-1}\,\mathrm{d}x$ $=0.86$	[M1] [A1]

5(i).	$\frac{1}{x}+\frac{1}{y}=\frac{1}{k}$ $\Rightarrow ky+kx=xy$ $\Rightarrow y(x-k)=kx$ $\therefore y=\frac{kx}{x-k}$	[B1]
(ii)	$y=\frac{kx}{x-k}=k+\frac{k^2}{x-k}$ $\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{k^2}{(x-k)^2}$ At, $(2k,2k)$ the gradient of the normal $=\frac{(2k-k)^2}{k^2}=1$ Equation of normal: $\frac{y-2k}{x-2k}=1\Rightarrow y=x$	[M1] [B1] [1] [A1]
(iii)	For the normal to cut the curve, $\frac{kx}{x-k}=x$ $\Rightarrow kx=x^2-kx$ $\Rightarrow x^2-2kx=0$ $\Rightarrow x(x-2k)=0$ $\Rightarrow x=0$ or $x=2k$ Since $x\neq 0$, the normal will not cut the curve again.	[M1] [B1] [A1]
6(i)	$P=\frac{1}{2}\pi r+2x+2r=200 \Rightarrow x=100-r-\frac{1}{4}\pi r$ $\therefore A=\frac{1}{4}\pi r^2+xr$ $=\frac{1}{4}\pi r^2+r\left(100-r-\frac{1}{4}\pi r\right)$	[B1] [B1]

	$=\frac{1}{4}\pi r^2+100r-r^2-\frac{1}{4}\pi r^2$ $=100r-r^2$	
(ii)	$\frac{\mathrm{d}A}{\mathrm{d}x}=100-2r$ $\frac{\mathrm{d}A}{\mathrm{d}r}=0,\ 100=2r\Rightarrow r=50$ r: 49.9, 50, 50.1 $\frac{\mathrm{d}A}{\mathrm{d}r}$: +, 0, - $\therefore$ *V* is maximum when $x=20$. When *A* is a maximum, $x=100-50-\frac{1}{4}\pi(50)=50-12.5\pi$ Maximum $A==100(50)-(50)^2=2500$	[B1] [B1] [A1] [A1]
7(i)	It would be impractical for the manufacturer to test every mobile phone produced. Therefore a sample is tested instead. **<u>OR</u>** any reasonable answer.	[1]
(ii)	Let the 3 models be model A, B and C in the ratio of 3:2:1. First determine the number of sample needed for each model. Take 60, 40 and 20 of model A, B and C respectively. Randomly select the necessary number of each model.	[1 - Correct numbers for each model, 1 - Random Selection.]
8(i)(a)	($\mathrm{P}(A)=\frac{45}{120}=\frac{3}{8}$ or 0.375	[B1]

(b)	$\mathrm{P}(A\cup B)=\frac{45+15}{120}=\frac{60}{120}=\frac{1}{2}$	[B1]
(c)	$\mathrm{P}(A\cap B)=\frac{20}{120}=\frac{1}{6}$	[B1]
(d)	$\mathrm{P}(A\vert B)=\frac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}=\frac{20/120}{20/120+15/120}=\frac{20}{35}=\frac{4}{7}$	[B1-intermediate; A1-correct answer]
(ii)(a)	$\frac{35}{120}\times\frac{34}{119}=\frac{1}{12}$ or 0.0833	[A1]
(b)	$\frac{35}{120}\times\frac{85}{119}\times 2=\frac{5}{12}$ or 0.417	[B1 if '2' is missing, otherwise B1, A1]
(c)	$\frac{20}{120}\times\frac{60}{119}\times 2=\frac{20}{119}$ or 0.168	[B1 if '2' is missing, otherwise B1, A1]
(iii)(a)	$\frac{35}{120}\times 0.96+\frac{85}{120}\times 0.08=\frac{101}{300}$ or 0.337	[B1-intermediate; A1-correct answer]
(b)	$\frac{\frac{35}{120}\times 0.96}{\frac{101}{300}\text{ or }0.33667}=\frac{84}{101}=0.832$	[B1-intermediate; A1-correct answer]
9	X = length of a rod ~ $\mathrm{N}(2,0.28^2)$ $H_0:\mu=2$ $H_1:\mu\neq 2$ Perform a two-tail test at 5% level of significance Reject H_0 if p-value < 0.05. Under H_0, $\bar{X}\sim\mathrm{N}\left(2,\frac{0.28^2}{150}\right)$ test statistic: $Z=\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}=\frac{\bar{X}-2}{0.28/\sqrt{150}}$	[B1- correct H_1] [B1- correct $\bar{X}$ distribution or test statistic]

	$\bar{x}=1.97,\ n=150,\ \sigma=0.28$ and p-value = 0.189. Hence H_0 is not rejected. There is no significant evidence at 5% level that a change in mean length has occurred.	[B1-correct p-value] [B1-correct conclusion]
(b)(i)	If the sample was not random there can be no confidence in the conclusion (or theconclusion is unreliable). For example, the sample may have been the first 150 rods produced and the mean length could have changed as production continues.	[B1]
(ii)	The conclusion is unaffected whether the distribution is normal or not as this is a large sample which means that the mean length of the metal rods is approximately normally distributed (central limit theorem).	[A1-correct answer + B1-reason]
(c)	$L<1.96$ or $L>2.04$ (allow $L<1.95$ or $L>2.05$)	[A1, A1]
10(i)	Let ***X*** denotes the number of rejected pens in a box of 250 $$X \sim \mathrm{B}(250,\ 0.03)$$ $P(X\geq 5)=1-P(X\leq 4)=1-0.128202=0.872$	[M1] [A1]
(ii)	Since n is large ($n>50$), $np>5$ and $nq>5$ $$X \sim \mathrm{N}(7.5,\ 7.275)$$ $P(X\geq 10)\xrightarrow{C.C}P(X>9.5)$ =0.22919	[M1] [1 for C.C, 1 for correct answer to desired accuracy]
(iii)	Let ***Y*** denotes the number of boxes of pen rejected in a single delivery. $$Y \sim \mathrm{B}(100,\ 0.22919)$$ $\mathrm{E}(Y)=100\times 0.22919=22.9=23$	[M1] [A1]
(iv)	Let $\overline{Y}$ denotes the mean number of rejected boxes per trip for the year. Since *n* is large, by the Central Limit Theorem,	[1,1] Only award marks

	$\bar{Y}=\frac{Y_1+Y_2+...+Y_{300}}{300} \sim N(22.919, \frac{17.666}{300})$ $P(\bar{Y}<23)=$ 0.63073=0.631	if students clearly stated CLT. [A1]
11(i)	<table><tr><td>y</td><td>8</td><td>5</td><td>7</td><td>3</td><td>15</td><td>8</td><td>7</td><td>13</td></tr></table>	[1 – 3 correct entries, 2 – 5 correct entries and 3- all correct.]
(ii)	Least square regression line of w on x $w=0.17661x+48.768$ $=0.177x+48.8$ When $x=18$, $w=0.17661(18)+48.768=51.9=52$	 [1] [A1]
(iii)	From the GC, $r=0.153$ The product moment correlation coefficient suggests that there is only weak positive correlation between the two variables; therefore a linear model may not be very appropriate. The estimate would not be reliable.	[A1] [1] Award zero if student just mention interpolation.
(iv)	The different x-values were scaled by different amount as the population of each area is different. The product moment correlation coefficient is only unaffected if the variable is scaled by the same amount.	[1]
(v)	From the GC, $r=0.950$ The product moment correlation coefficient suggests that there is very strong positive correlation between the two variables; therefore a linear model is appropriate.	[A1] [1]
(vi)	The variable *W* and *Y* are both dependent on *Z* while *X* is independent of *Z*. Thus, looking at the data for *W* and *X* may not give an accurate	

	picture of any relationship between tertiary education and level of household living in condominiums or landed properties.	[1]
(vii)	No, the answer in (v) does not lead to the given conclusion as correlation does not imply a cause and effect relationship.	[1]
12(i)	Let ***X*** denotes the diameter of a randomly chosen football. $X \sim \mathrm{N}(22,\ 0.09)$ $P(21.7 < X < 22.3) = 0.68269 = 0.683$ $P(X < 22) = 0.5$ $\text{Probability} = 3 \times 0.68269 \times (0.5)^2 = 0.512$	[B1] [B1] [A1]
(ii)	$P(22 - a < X < 22 + a) = 0.75$ From the GC $22 - a = 21.65490$ and $22 + a = 22.34510$ $\therefore a = 0.345$	[1] [A1]
(iii)	Let ***Y*** denotes the diameter of a randomly chosen basketball. $Y \sim \mathrm{N}(24,\ 0.16)$ $\overline{X} = \dfrac{X_1 + X_2 + \ldots + X_{10}}{10} \sim N(22, \dfrac{0.09}{10})$ $\overline{Y} = \dfrac{Y_1 + Y_2 + \ldots + Y_5}{5} \sim N(24, \dfrac{0.16}{5})$ $\overline{X} - \overline{Y} \sim N(-2, \dfrac{0.09}{10} + \dfrac{0.16}{5})$ $P(-1.5 < \overline{X} - \overline{Y} < 1.5)$ $= 0.00677$	[1-correct mean and variance] [M1-expression] [A1]
(iv)	We assumed that the diameter of the football and basketball chosen follow independent normal distributions.	[1]

(v)	Diameter of basketball hoop $= \frac{78}{\pi}$ $P(Y < \frac{78}{\pi})$ $= 0.98079$ $= 0.981$	[1] [M1] [A1]